

EEG Based Autism Detection Using CNN Through Correlation Based Transformation of Channels' Data

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Abstract—Early diagnosis of autism or autism spectrum disorder (ASD) can help improving behavioral, language development and communication skill. As ASD is a neurodevelopmental disorder, brain signals are used to early diagnosis. Among different brain signals, electroencephalography (EEG) is the effective one. Electrical brain activity is measured through EEG signal from the scalp of human over a period of time and is used to analyze complex neuropsychiatric problems of brain. This study investigates an EEG based ASD detection using CNN, the well-known deep learning method for image analysis and classification. At first, the individual EEG channel data are transformed into 2D form using Pearson's Correlation Coefficient and then classification is carried out using the well-known CNN model residual neural network. Experiments performed on clinical EEG data show that the proposed approach achieved a classification accuracy of 100%.

Keywords— Autism Spectrum Disorder, Convolutional Neural Network, Electroencephalography (EEG), Pearson's Correlation Coefficient, Residual Neural Network.

I. INTRODUCTION

Autism spectrum disorder (ASD) is related to complex many factors brain development leading to social and communication impairments, and restricted stereotyped behaviors [1]. ASD happens in youngsters and keeps on growing to adolescences and grown-ups. In the majority of cases, side effects of ASD show up within the initial 5 years of life[2]. Now-a day's ASD detection is an important issue as it affects the physical and mental growth in children. Prevalence of ASD one out of every 68 children has been reported in many studies in the last few decades but recent studies show that one out of every 59 children is diagnosed with ASD [3]. Other surveys worldwide say that the number of ASD incidence is rising globally [4]. Fig. 1 shows an autistic child's situation. There is no way of preventing ASD, but there are treatment options. There exists increasing evidence that autism may not necessarily be lifelong. Early diagnosis and intervention is most helpful and can improve behavioral and communication skills and language development [5], [6].

There are various approaches to diagnose ASD. ASD is described by tireless deficiencies in social correspondence with others symptoms[7]. Some techniques detect ASD through a questionnaire-based screening method that provides only suggestions for additional investigation. Some methods work

with a multidisciplinary team that diagnoses ASD interacting with patients.

ASD is also diagnosed based on brain activity through different brain signals as its cause is the neurodevelopmental disorder. Among different brain signal methods, electroencephalography (EEG) is being increasingly explored as a potential clinical tool for monitoring brain activities as it is relatively easy-to-use as well as low cost [8]. EEG quantifies the electrical signs of the brain by means of electrodes that are set on different places on the skull. In examining the human brainwave to perform data grouping on the got signals, EEG has been demonstrated to be a compelling medium [9].

Recently, few studies on early ASD detection using different machine learning (ML) methods from EEG are reported [8], [10]–[13]. Grossi et al. [13] used a new EEG preparing algorithm named MS-ROM/I-FAST alongside different ML based algorithms to characterize autistic patients. EEG signal was used in [14] to classify autistic child from normal ones through Artificial Neural Network (ANN). Jamal et al. [15] diagnosed existence of ASD utilizing the operative cerebrum availability estimates obtained from EEG signal of children through face recognition assignments. Classification was performed through Linear Discriminant Analysis (LDA) and Support Vector Machine (SVM) achieving an accuracy of 94.7%. EEG analysis of autistic children of different conditions based on potential differences between children was carried out in [11] using various models such as empirical mode



Fig. 1. A child with Autism. Source: www.eparent.com

decomposition (EMD) and second-order difference plot (SODP) models. An efficient EEG signal based ASD diagnostic approach was proposed in [20]. They applied signal processing techniques on collected subjects and classification was performed using ANN and SVM. Heunis *et al.* [16] investigated Recurrence Quantification Analysis (RQA) as a biological marker for autism through EEG data.

This study investigates EEG based ASD detection using Convolutional Neural Network (CNN). CNN is the prominent deep learning method for complex perceptual tasks especially, analysis and classification of image data such as hand written numeral recognition [13]. In the proposed method, firstly, individual EEG channel data are transformed into 2D form using Pearson's Correlation Coefficient [19]. Finally, classification is performed employing the image like 2D data using the well-known CNN model residual neural network (ResNet).

The structure of the paper is as follows. Section II clarifies the proposed methodology in detail. Experimental results and discussion are presented in Section III. Finally, Section IV concludes the paper with brief summary of the study.

II. METHODOLOGY

The main contribution of this study is the transformation of EEG channel data to 2D representation and visualization using PCC. Schematic of the proposed method is shown in Fig. 2. There are two function steps of the method. At first, EEG data is transformed to 2D image using PCC. And then, CNN is used to classify the image into two categories: ASD and control (i.e., non ASD). Following subsections briefly describe EEG data preparation and two functional steps, 2D transformation of EEG data and finally, classification using CNN.

A. EEG Data Preparation

EEG encompasses different waveforms of diverse amplitude, frequency, spatial distribution and reactivity to different stimuli. Analysis of EEG is performed on its different frequency bands fluctuating the bandwidths. EEG has been demonstrated to be a compelling medium to consider human brainwave so as to represent information arrangement on the acquired signals [9]. Universal 10–20 framework [17] is utilized to portray and apply the area of scalp cathodes with regards to an EEG test. Fig. 3 shows the schematic representation of international 10-20 system of electrode placement of EEG recording. The anteroposterior measurements are based on the distance between nasion and inion over the vertex in the midline with five category points:

central (C), frontal pole (Fp), frontal (F), parietal (P) and occipital (O). To differentiate between homologous positions on left and right side of hemispheres, it has been decided to use even numbers as subscripts for the right hemispheres and odd numbers as for the left hemisphere. As an example, the first point of Fp (i.e., Fp1) is 10% of nasion-inion distance from the nasion. The data from individual channels are recorded and stored in computer system.

B. 2D Transformation of EEG Data

Pearson's Coefficient of Correlation (PCC) or bivariate relationship is a proportion of a straight relationship between two factors of one measurement. It is, numerically, the covariance proportion of two exhibits to the aggregate of their standard deviation. The value 1 suggests between -1 and + 1. Where + 1 implies a 100% positive correlation, 0 does not indicate a correlation, -1 means a 100% correlation but an anti or negative correlation. The correlation coefficient of Pearson is usually defined by the Greek letter ρ . Because of a couple of random variables (X, Y) , the formula for ρ is described by Eq. (1).

$$\rho_{X,Y} = \frac{cov(X,Y)}{\sigma_X \sigma_Y}, \quad (1)$$

where, cov is the covariance, σ_X and σ_Y are the standard deviation of X and Y respectively.

The images include the attributes of two consecutive channels dataset of the PCC. Firstly, the PCC is determined between the combination of each two channels to formulate the images. The coefficients are later arranged in a matrix form, and

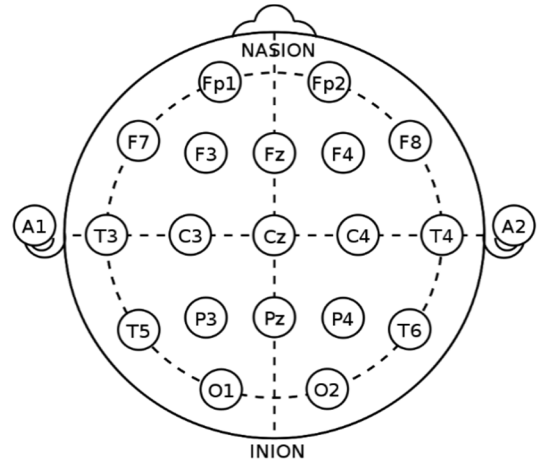


Fig. 3. International 10-20 system electrode placement for EEG recording.

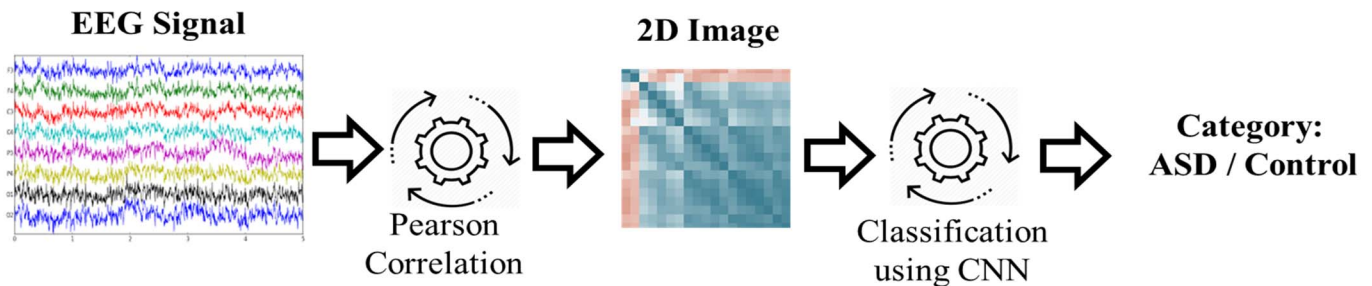


Fig. 2. Schematic of the proposed method.

a data array can be found. In this paper, since data of 19 EEG channels are considered, the size of the PCC matrix will be 19×19 . The matrix was plotted as a color image using the seaborn heatmap library from Python. Examples of this type of formulated color images are shown in Fig. 4 for two sample data. The representation clearly distinguishes ASD matter in Fig. 4(a) from control (i.e., non ASD) matter in Fig. 4(b).

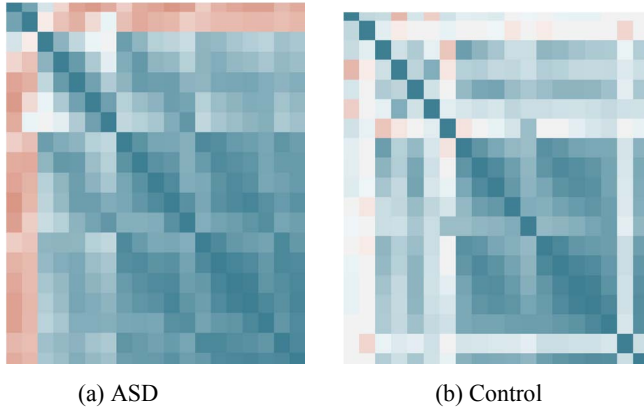


Fig. 4. Sample 2D images generated from correlation matrix.

C. Classification Using CNN

A CNN is a deep learning algorithm equipped for catching an information picture, relegating essentialness (learnable weights and biases) to various parts of the picture and having the option to recognize them. CNN requires less pre-processing compared with other classification algorithms. While the filters are hand-built in simple procedures, CNN can become familiar with these qualities with adequate training [20]. This study considered ResNet as the well-known CNN model for classification of EEG data transformed into 2D image.

ResNet is a residual learning framework which facilitates the training of substantially deeper networks. The layers are specifically reformulated by comparison to the layer inputs as learning residual functions, rather than learning unreferenced functions. At the beginning of each level, a convolutional layer with steps of 2, halves the feature map size while the number of filters doubles. The layers have the same number of filters inside each level. A ResNet building block is called a residual block, or block of identity. A residual block is simply when a layer is quickly redirected to a deeper layer inside the neural network. A fundamental description and implementation issues of ResNet are available in [21].

III. EXPERIMENTAL STUDIES

A. Experimental Data

The EEG data used in this study were collected from Villa Santa Maria Institute, Italy [13]. EEG data were recorded with the enlisted subjects at resting state. There were autistic children of seven to fourteen years of age (average score = 10.4) and the normal group was made out of four boys and six girls of 7 to 12 years of age (normal = 9.2). A Micromed device

provided by SystemPlus Evolution software, using pre-wired headsets with cotton elastic inside from nineteen electrodes was used for EEG recordings with bandpass filter 0.3–70 Hz. Electrodes were situated by the International 10–20 System. Fifteen youngsters and teenagers (13 men; 2 women) got an autonomous finding for ASD as indicated by DSM-V principle, which was then affirmed by a certified kid and juvenile specialist at Villa Santa Maria.

B. Experimental Setup

The algorithm is implemented on anaconda. The experiments have been done on a PC (Intel® Core™ i7 - 4720HQ CPU @ 2.60 GHz, 8 GB RAM) with Windows 10 OS. In the train-test protocol, a train with 17 cases and a test with 8 cases were performed. The Resnet uses the following libraries: Keras, NumPy, Os, Pandas, Sklearn, Matplotlib, Pyplot and Tensorflow.

C. Experimental Results

Figure 5 shows classification accuracy on both the training and test sets while number of training epochs varied up to 200. From the figure it is observed that training accuracy reached 100% after 25 epochs. On the other hand, test set accuracy also reached 100% after 160 epochs which is promising. It is notable that previous study also achieved the same performance [13] on the same data.

Table I compares ASD detection classification accuracy of the proposed method with other existing methods on clinical EEG data. The methods used by different studied and dataset size are also presented in the table. Existing methods considered ANN, LDA, SVM, Multi-layer Perceptron NN (MLP), Layered Recurrent NN (LRN) and Principle Component Analysis (PCA). It is already mentioned that the data of [13] is used in the present study. Therefore, comparison with other methods excluding [13] is not completely justified due to different data uses. However, the proposed method outperformed any other method. The proposed method achieved 100% accuracy like that of the performance of the data owner's experiments. Finally, the significance of this study is the innovation in the development of 2D image representation of EEG data which revealed as a promising method for ASD detection from EEG data.

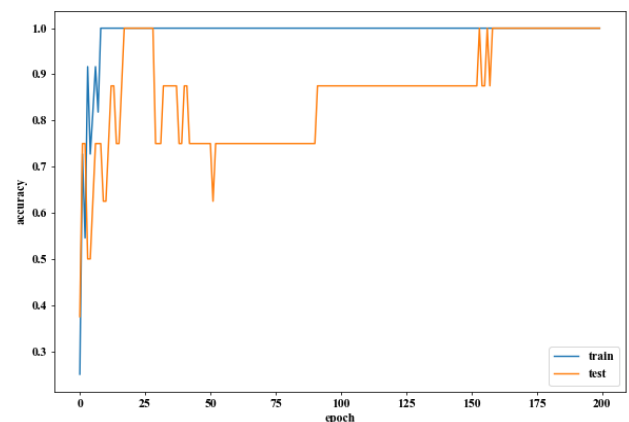


Fig. 5. Classification accuracy of ResNet varying training epochs.

TABLE I. CLASSIFICATION ACCURACY COMPARISON.

Sl.	Work Ref, Year	Sample Size (ASD + Control)	Method Used	Accuracy
1	Raja and Priya [14], 2017	10 (4+6)	ANN, LRN	94.62%
2	Harun and Hamzah [22], 2018	-	ANN, SVM	90.05%
3	Heunis and Aldrich [16], 2018	111 (39+72)	MLP, SVM, LDA	92.9%
4	Hadous and Alafeef [11], 2019	36 (18+18)	PCA, ANN	97.2%
5	Grossi and Oliveri [13], 2017	25 (15 + 10)	ANN	100%
6	Proposed Method	25 (15 + 10)	PCC, CNN	100%

IV. CONCLUSIONS

As ASD is a neurodevelopmental disorder, diagnosis based on brain activity is considered as the most effective for early detection. Among exiting brain signal methods, EEG is a relatively easy-to-use as well as low cost, therefore, it is promising for ASD detection. In this study, EEG-based detection of ASD is performed by representing individual channels' data in 2D form and then classifying it with ResNet of CNN. A potential extension of the present study is to investigate the method on datasets having large number of data.

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